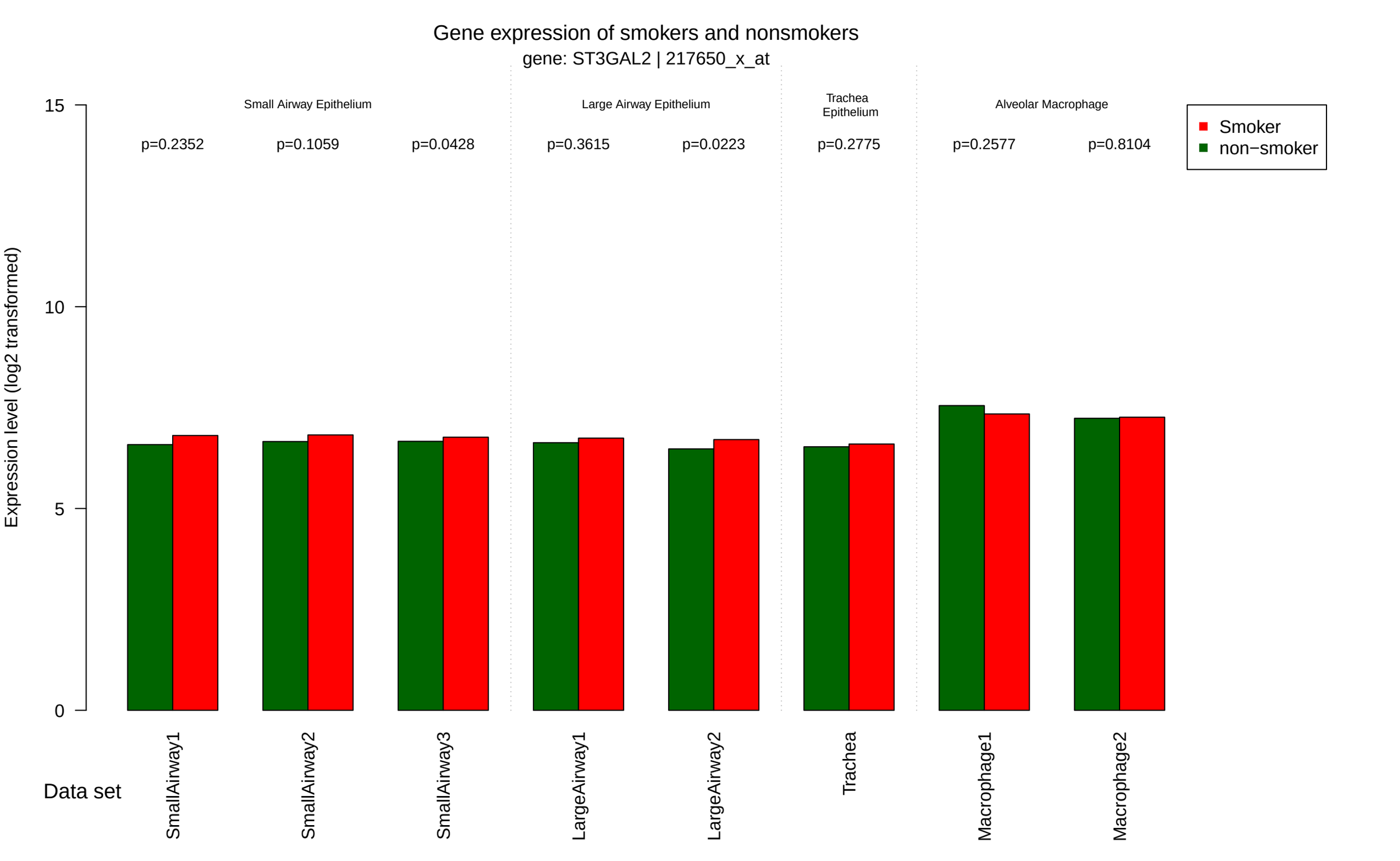
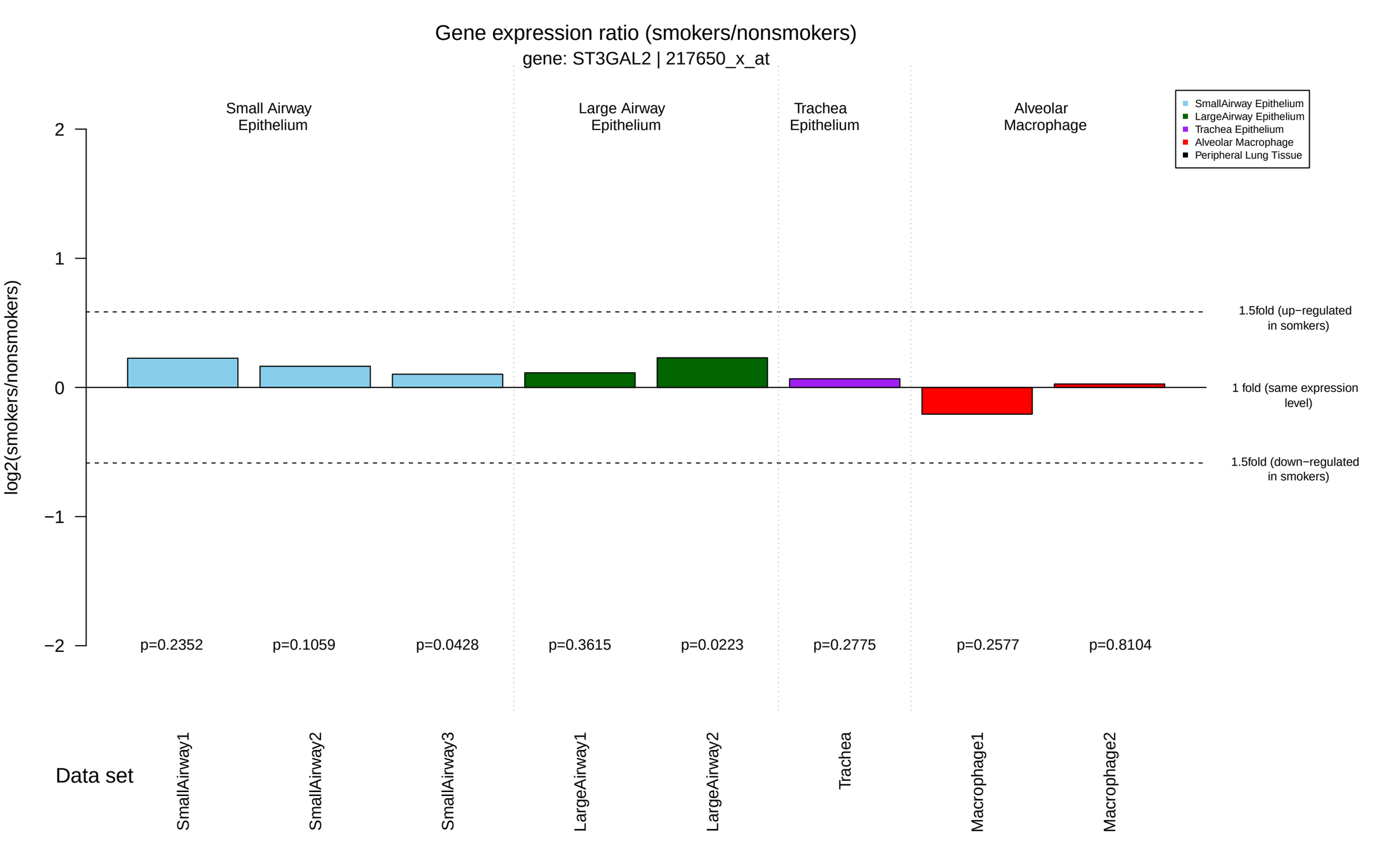
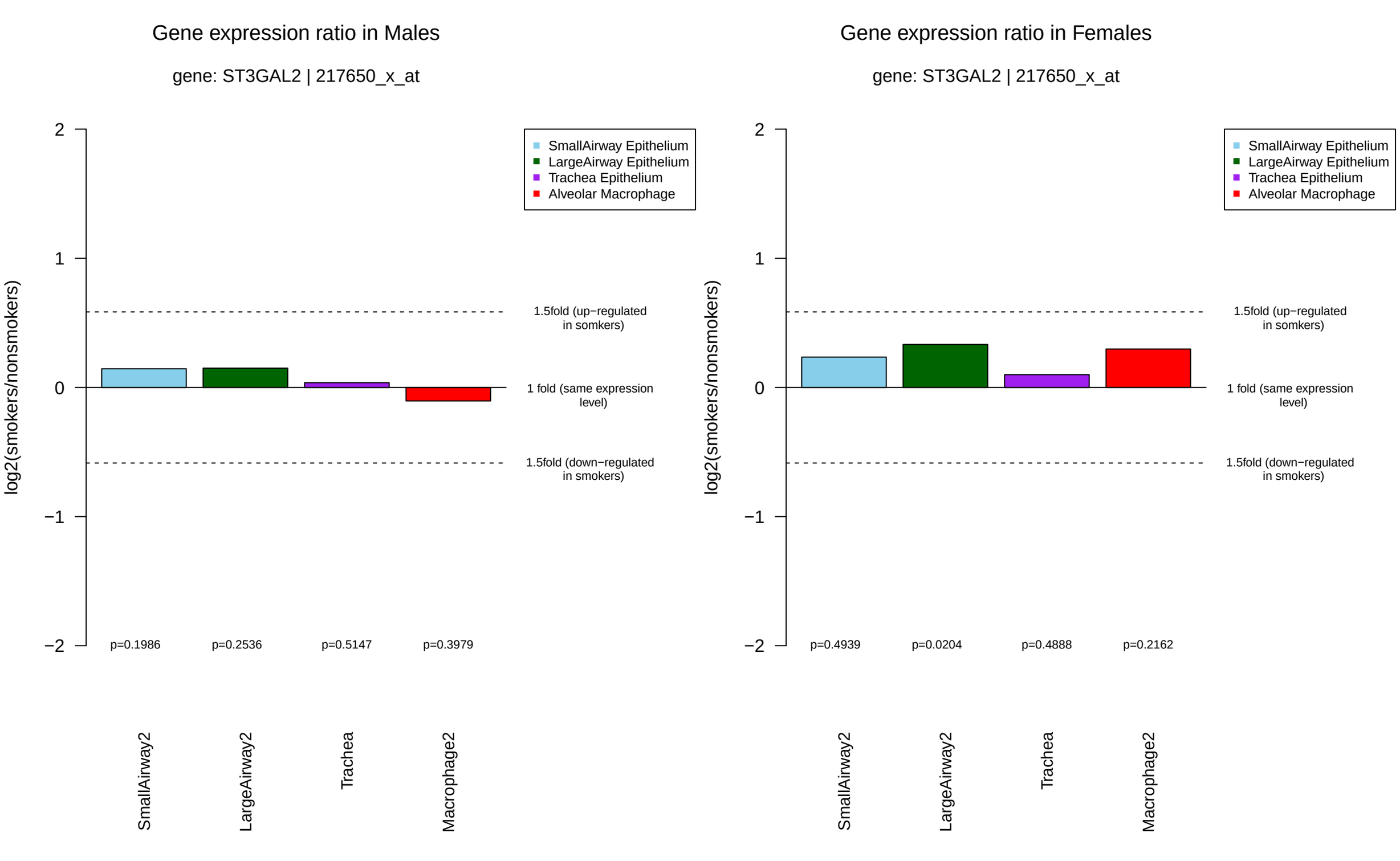
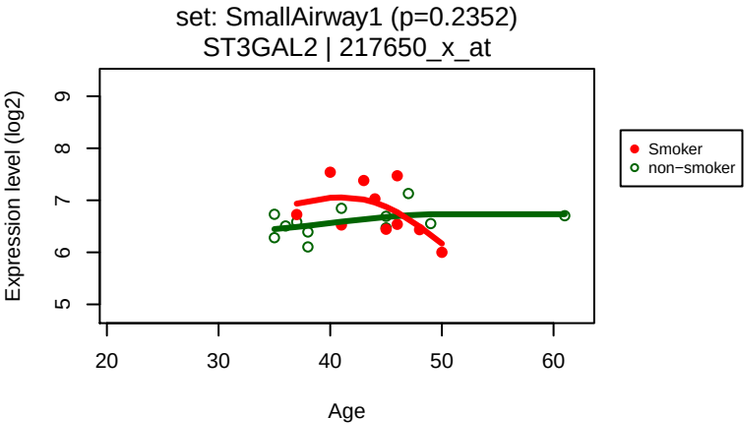
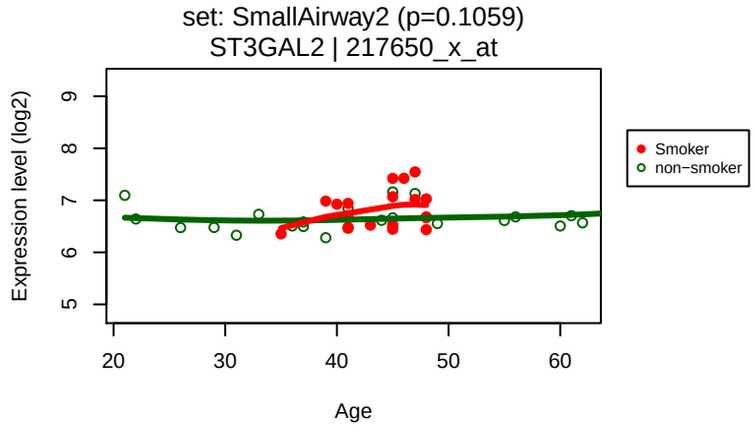
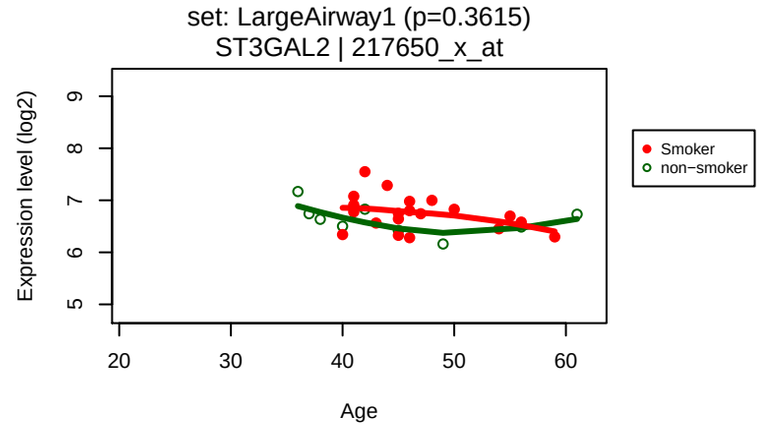

Gene expression of smokers and nonsmokers
gene: ST3GAL2 | 217650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2352
p=0.1059
p=0.0428
p=0.3615
p=0.0223
p=0.2775
p=0.2577
p=0.8104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

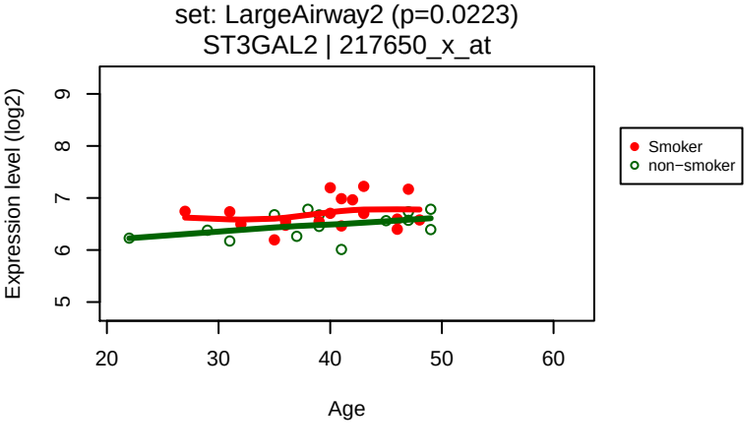

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL2 | 217650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2352
p=0.1059
p=0.0428
p=0.3615
p=0.0223
p=0.2775
p=0.2577
p=0.8104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

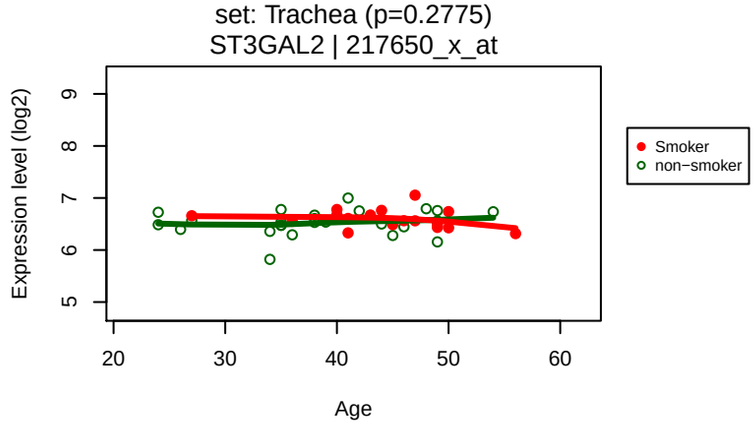

Gene expression ratio in Males
gene: ST3GAL2 | 217650_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1986
p=0.2536
p=0.5147
p=0.3979
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL2 | 217650_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4939
p=0.0204
p=0.4888
p=0.2162
SmallAirway2
LargeAirway2
Trachea
Macrophage2

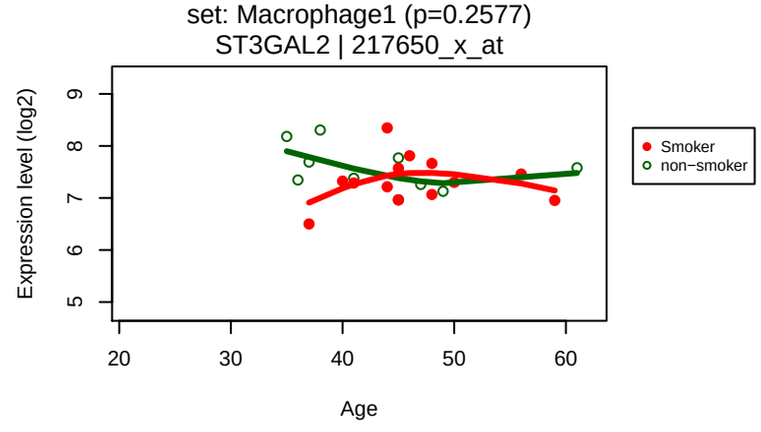

set: SmallAirway1 (p=0.2352)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker

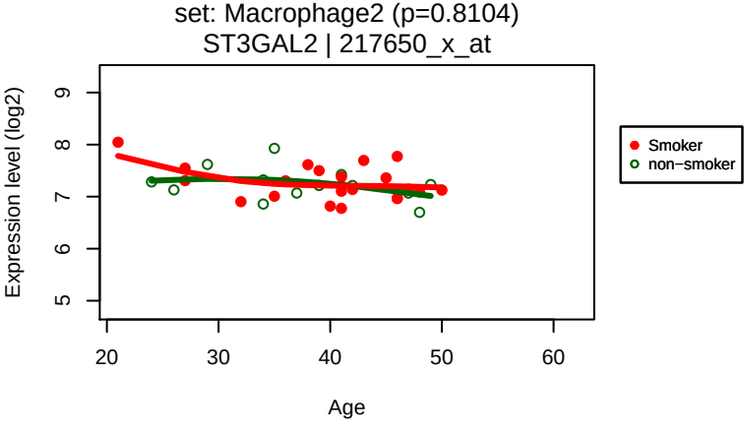

set: SmallAirway2 (p=0.1059)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker

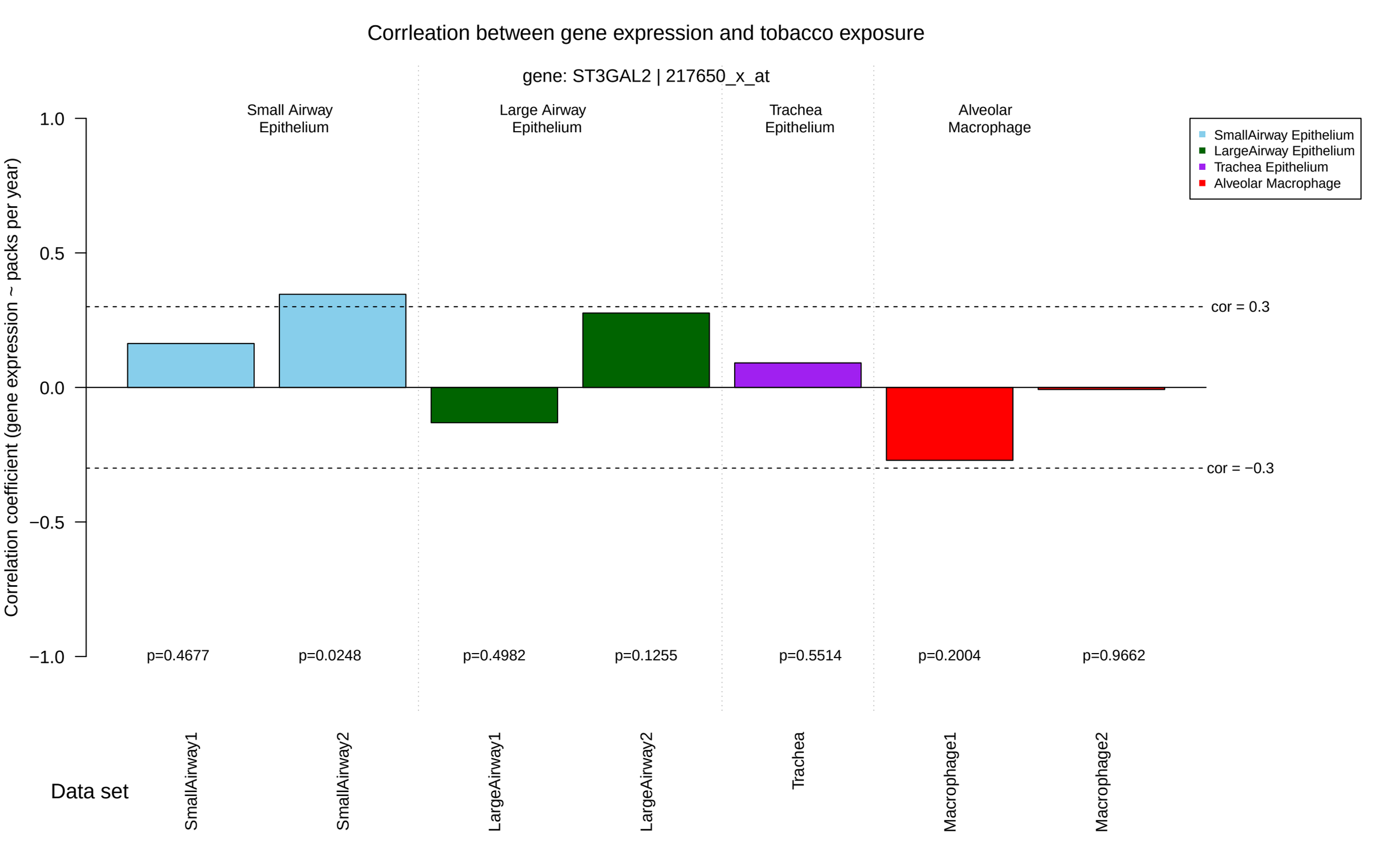

set: LargeAirway1 (p=0.3615)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0223)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2775)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2577)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8104)
ST3GAL2 | 217650_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL2 | 217650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4677
p=0.0248
p=0.4982
p=0.1255
p=0.5514
p=0.2004
p=0.9662
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2